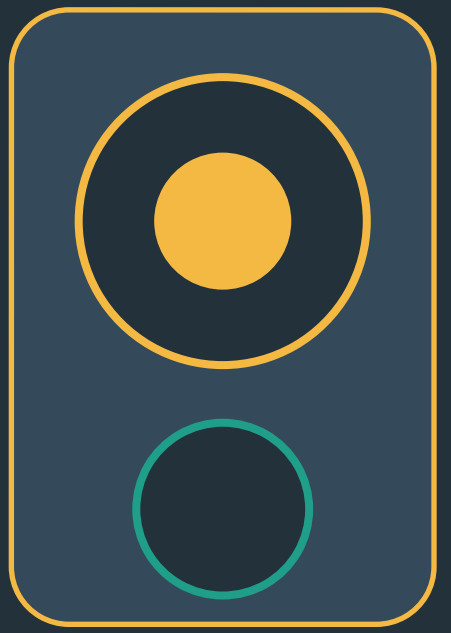
AI FOR FOUNDATIONAL LEARNING HACKATHON 2026 · BUILD STAGE

Sahpaathi

A classroom companion in a ₹500 speaker.
The teacher works closely with one group; Sahpaathi keeps another engaged, in the children's home language, with no internet.



Live demo sahpaathi.vercel.app
Source github.com/Harsha-code-per/Sahpaathi
Challenge Classroom Complexity (1A language, 1B multigrade)
Team A1pha
Test accounts and sample inputs: last slide

WHAT WAS BUILT

The Ideate plan is now a working classroom loop

Write

Claude writes a structured audio activity for a chapter and a district: pauses, cues, three branches.

Check

Code, an AI reviewer and back-translation check it. A human approves every line.

Voice + pack

Approved lessons are voiced and signed into a pack of about 1 MB.

Teach offline

One tap to start. Most / Some / Few picks the next branch on the phone.

Follow up

Absences queue catch-up calls. Taps reach the block dashboard on next signal.

What changed from the Ideate deck, and why

Ideate deck	What shipped	Why
“AI co-teacher”	“Classroom companion”; same loop	Education mentor: claim engagement, and support the teacher
Kotlin / React Native app	Installable web app (PWA)	A link installs it: no app store, and evaluators can open it
Bhashini, IndicTrans2, Indic TTS	Sarvam Bulbul v3 voices	Tamil, Hindi, Odia and English from one API
RAG over textbooks (pgvector)	Curated chapters + NIPUN outcomes	8 chapters needed accuracy, not search
22 languages, Santali to Odia showcase	Lessons in Tamil, Hindi, English; Odia app text and voice ready	No Odia lesson yet: it waits on Santali, with 0 of 42 phrases recorded
Free call-back on real phones	Full IVR flow, tested in code	Not live: needs a voiced pack (voice credits ran out) and an Exotel number

AI runs once per lesson in the cloud; the classroom replays it

Layer	What happens	AI or code	Where in the repo
Data input	Chapter text, NIPUN outcome, district profile (languages, names, crops, free props), class taps	No AI: curated data	content/
Processing	Writes the lesson as schema-bound JSON: parts, pauses, cues, easier / core / stretch, analogy, call recap, chalk sketch	Claude Sonnet 5.5	backend/app/factory/
Guardrails	Rules and arithmetic, then rubric review with one auto-repair, then blind back-translation, then human approval	Code + Claude Haiku 5.5 + reviewer	factory/, Studio
Evaluation	Scores saved per lesson; 20 planted-defect test; in class, % Most got it per part; call completion	Code + Haiku	eval_guardrails, /insights
Output	Voice normalised to -16 LUFS, signed pack, offline player, phone recap	Sarvam Bulbul v3, ffmpeg, Ed25519	voice/, app/src/teacher/, ivr/

Other LLM calls

- Feedback loop: Claude rewrites the part classes struggled with, and it goes back through review
- Teacher recordings: Sarvam speech-to-text, then Haiku gives tips on clarity, pace and safety
- Parent helper prototype at /tutor: Sonnet plans a topic, Haiku judges answers and phrases replies

Why this design

No LLM runs in the classroom, so nothing there needs a network, and a mistake cannot reach children live.
Branching is plain code on the phone. In the parent helper, code rules decide pass, hint and reteach: the model does not.

Four guardrails stand between an AI draft and a child

1 Code

Schema, rule checks, every sum recomputed, language handover checked

2 AI rubric

Haiku scores pedagogy, age fit, accuracy, language bridge, localisation and safety; one auto-repair

3 Back-translation

A blind English translation sits beside every line, so a reviewer can catch meaning drift

4 Human

A native speaker edits lines and approves. Nothing is voiced until then. The AI's original is kept

MEASURED ON PLANTED DEFECTS

18 / 20 caught and pinpointed by code or the AI reviewer

20 / 20 caught by any tier, before a human looks

Wrong sums, beyond-grade maths, unsafe instructions, dropped home language, answers given too early. Small sample: 3 lessons, one run (21 Oct).

Beyond the lesson

- Packs are Ed25519-signed with sha256 hashes: the app refuses a tampered pack
- Audio QA: each line within 1.5 LU of -16 LUFS, none over 45 s, pack under 2 MB
- Choral mic only suggests a branch; the teacher decides. No tap in 10 s falls back to Core
- No child names or phone numbers: classrooms and roll numbers are opaque codes
- Paid and public endpoints have daily caps stored in the database, returning HTTP 429
- Honest limit: the parent helper prototype does not pass through the four lesson guardrails; camera sensing is off and has not been tested on real faces

Built for no signal, little data and a ₹7,000 phone

Part	Needs network?	What happens offline
Player, branching, analogy, attendance, mic hint	**No**	Works fully in airplane mode after one visit
Taps and attendance sync	**Any signal**	Queued on the phone; retried every 30 s, sent in one small request
New lessons	**Any signal, resumable**	A dropped download resumes; packs also pass by file share or SD card
Catch-up call	**Voice network only**	The parent needs no data or smartphone; the call waits for the sync
Studio, Admin, lesson generation	**Yes (desk tools)**	Not on the classroom path
Parent helper at /tutor	**Yes (prototype)**	App warns and points back to the offline classroom

0.5 s
repeat load on Slow 3G with 4× slower CPU (lab)

96 KB
first-visit transfer, 6.6 s to lesson list (lab)

~1.1 MB
a voiced lesson of about 10 minutes, opus 16 kbps

100
Lighthouse accessibility, mobile

Hardware: any Android with Chrome, a Bluetooth or AUX speaker, any keypad phone at home. Also run on a moto g34. Still to do: the 3-metre speaker test in a real room, and a ₹7,000 phone.

An education mentor's review reset our positioning

40-minute live demo with an education mentor from Central Square Foundation, 21 Oct. Findings logged in docs/TESTING_LOG.md.

What we heard	What we did	Where it is
“Second teacher” overclaims: in a noisy room a speaker alone may not solve active learning	Now a classroom companion: it helps the teacher manage groups, and does not replace her	Landing page, README, deck, video
You can claim engagement at this stage, not that learning happened	We claim engagement only. No learning gain is claimed	README, Evidence page
How does it look in action? The direction must reach the teacher	“Set up the room” before every lesson: teacher with those who need her most, one child as helper, in 4 languages	Teacher app, lesson prep
Grade 1–3 children cannot run the study mode alone; it can serve parents	/tutor is now a labelled prototype parent helper, off the home hero, with a short spoken note for parents	/tutor, landing page
Few features that work beat many. Is the script engaging for a Grade 2 child?	Effort moved back to the classroom loop: five activity types (pebble, slate, pair work, song, story) and teacher controls	Built-in lessons, class settings
Indic models err at scale; launch with a data caveat. Use state teacher guides as input	Not done yet: on the roadmap. Native-speaker sign-off stays a required guardrail	Roadmap

Honest status This is one expert review, plus the team's own device tests. No teacher or child has used it yet in a real classroom, and no learning outcome has been measured. Teacher sessions with written consent are the first roadmap item.

Every test that failed changed the product

Date	Test	What we saw	What we changed
19 Oct	Airplane mode, then back online	Queue waited for an app reopen: the only sync attempt ran before data flowed	Retry every 30 s (PR #43). It then drained 20 s after the API returned
19 Oct	Voice QA on 3 real packs	2 of 3 failed on one quiet line (-18.8 and -17.6 LUFS)	Gain re-measured on the encoded line; all 3 now pass
21 Oct	Clicking through the live site	White screens after deploys, and a crash on sending a tutor message	Fixed; every PR now runs a browser test of each page, plus accessibility
25 Oct	moto g34: full lesson	A lesson needs pause, a screen that stays on, and protection from a mistaken back press	Pause and resume, screen wake lock, confirm before leaving
1 Nov	Pre-submission audit of live flows	The live site played no sound: packs failed on a rejected voice key, and built-in lessons were text only	New voice key; packs rebuilt; the five built-in lessons now speak offline; recordings allowed through the body limit

Metrics promised in the Ideate deck 0 MB in the classroom: shown. +20 guided minutes, 80% Most got it, 60% catch-up calls completed: not measured yet, because the school pilot has not happened. They move to the pilot.

These are team and device tests, not classroom results. Automated: 272 backend and 229 app tests, plus browser smoke tests on desktop and phone, on every pull request.

Next: from a working loop to proof in real classrooms

Next 4 weeks

- Teacher sessions with written consent, using our interview guide and log; test whether Grade 2 children can follow the instructions
- Native-speaker review of Tamil, Hindi and Odia lines and voices; compare scripts with Pratham's audio and SCERT teacher guides
- Run the 3 m speaker test on a real ₹7k phone
- Connect Exotel and make the first real catch-up call

Next 3 months

- Pilot 10 to 20 multigrade schools with one DIET or Block Resource Centre
- Measure engaged minutes per child, % Most got it after branching, call completion; only then test learning
- Feed SCERT teacher guides into generation; collect recorded data with a rubric to improve voices
- Add literacy, aligned to NIPUN Bharat

Languages

- Scheduled languages through Indic TTS, each with a native reviewer
- Tribal languages (Santali first, then Gondi, Bhili, Kui) through consented community phrase banks; the splicer is built

Parent helper (prototype)

- Add the four guardrails to its answers
- Verifiable parental consent (DPDP Act 2023) and legal review before any child uses camera sensing
- Real-data training only from consented pilots

It plugs into systems that exist, at a few rupees per child

Existing system	How Sahpaathi fits
NIPUN Bharat, Samagra Shiksha	Lessons tagged by learning outcome; dashboards roll up to block and district
CRPs	Already visit monthly: carry packs by phone or SD card, and see the dashboard
DIKSHA	Publish approved lessons and chalk-sketch cards as open content
State IVR, community radio	The same audio plays on existing lines and slots

PER CHILD PER YEAR, MODELLED

₹6 without calls, at 1,00,000 classrooms

₹93 with a call for each absence

A lesson costs about ₹205 once. Calls are about 90% of cost at scale, at an estimated ₹1 a minute; confirm with Exotel's rate card.

Adoption path

1. **One DIET or BRC** names schools and reviewers
2. **Reviewer approves** lessons per language and grade
3. **CRP carries packs** to schools with no signal
4. **Block officer sees** which lessons and classes need help

FOR EVALUATORS

How to test it: accounts, five sample inputs, offline steps

Role	Open	Email	Password
Teacher	/teacher (no login to play)	teacher@sahpaathi.test	sahpaathi#01
Reviewer	/studio	reviewer@sahpaathi.test	sahpaathi#01
Admin / block officer	/admin	admin@sahpaathi.test	sahpaathi#01
Parent helper	/tutor (no login)	none	none

Test it offline

1. Open /teacher in Android Chrome
2. Add to Home screen (optional)
3. Wait for a few seconds, then turn on airplane mode
4. Open a lesson and tap Most / Some / Few

Five sample inputs

1 Teacher	Open “Counting to three with pebbles”, read Set up the room, Start. Listen, tap Few got it, then Most got it at the next check. Let one check run out: it continues after 10 s.
2 Teacher	Play “The thirsty crow” (Grade 2 story, Tamil and English). After the first page loads, turn on airplane mode: the voice still plays, and the end screen asks who was absent.
3 Reviewer	In /studio open Approved, then the Grade 3 Madurai lesson. Read each Tamil line beside its English back-translation and the six AI scores. “Write a new lesson” starts Claude (slow on the free server).
4 Admin	In /admin open Overview and Classes: the class-health board, marked SAMPLE. Lesson packs and Catch-up calls need a voiced pack, which is not built on this demo (voice credits ran out).
5 Parent helper	In /tutor turn on Study and type “place value of two-digit numbers”. Answer one question wrongly, then say “explain in Hindi” and “I am tired”.

TEAM

A1pha



Harshavardhan K

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Member

Sahpaathi: a classroom companion for the group the teacher can't reach right now.